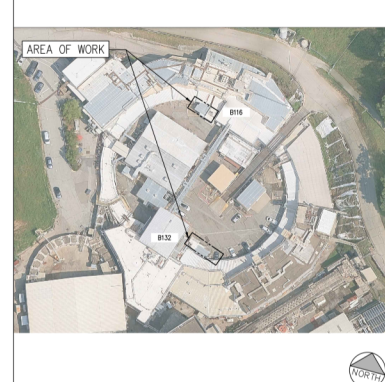


B116 AND B132 HVAC DESIGN

AREA OF WORK

[illegible]

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: onslater	
DATE: 1/23/2026	FRSE: 25-254
Revised Authorization	☒ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and IBC program requirements.	
This authorization of these plans shall not be construed to be a permit or an approval for violation of any applicable codes, standards, or IBC programs.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

THESE PLANS SHALL BE KEPT ON THE JOB SITE
FOR ALL CONSTRUCTION INSPECTIONS.

vektor Engineering & Consulting Services, Inc.
"Where engineering and technology drive innovation"
 2603 Camino Ramon, Suite 417
 San Ramon, CA 94583
 +1 (866) VEKTOR1 (835-8671)



SCALE: AS NOTED	DO NOT SCALE DRAWING	S5010.2025.01
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U.S. DEPARTMENT OF COMMERCE

U.S. DEPARTMENT OF ENERGY
ENERGY

THE DRAWINGS, SPECIFICATIONS AND OTHER DATA HEREIN PROVIDED ARE NOT TO BE COPIED, PUBLISHED OR OTHERWISE FURTHER DISSEMINATED.

PRIOR WRITTEN PERMISSION OF STANFORD UNIVERSITY / S	
DATE	ASSIGNABLE

ENGR	KOROLEV, SERGEY	12-10-2025	APPROVALS
DLS	FERNANDEZ, HUGO	12-10-2025	BO-OPRA, ANDI

	B116 AND B132
	B116 AND B132 HVAC DESIGN
	GENERAL
	COVER SHEET

DATE 10-20-24	DRAWING NUMBER B0116 C00 0001	REV
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B0116-G00-0001 - D	E
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GENERAL NOTES

1. ALL WORK SHALL CONFORM TO THE 2022 CALIFORNIA MECHANICAL CODE.

2. THE SUBMITTAL OF A BID PROPRIAL SHALL BE CONSIDERED CONCLUSIVE EVIDENCE THAT THE CONTRACTOR IS THOROUGHLY FAMILIAR WITH THE INTENT OF THE CONTRACT DOCUMENTS. THE CONTRACTOR PRIOR TO BIDDING, SHALL CHECK SITE CONDITIONS, INSTALLATIONS AND LISTINGS RELATED TO EXISTING UTILITIES, INCLUDING BUT NOT LIMITED TO, ELECTRICAL AND MATERIAL REQUIRED FOR COMPLETION OF THE SYSTEM.

3. CONTRACTOR SHALL BE RESPONSIBLE FOR HAVING VISITED THE SITE AND SATISFIED HIMSELF THAT THE CONTRACTOR'S UNDERSTANDING OF THE WORK WILL BE PERFORMED. THE CONTRACTOR SHALL CHECK ALL EXISTING CONDITIONS WHICH MAY AFFECT HIS WORK. NO ALLOWANCE WILL BE MADE FOR ANY EXTRA EXPENSE WHICH MAY BE INCURRED DUE TO NEGLIGENCE OR FAILURE TO EXAMINE THE SITE. EXISTING CONDITIONS, UTILITIES, ETC. THE SITE VISIT MUST BE DURING THE BID WEEK AND/OR PREPARED WITH THE OWNER.

4. CONTRACTOR SHALL PROVIDE QUALIFIED TRADE PROFESSIONALS WITH ADEQUATE SAFETY TRAINING IN CONSTRUCTION OF THESE TYPES OF ACTIVITIES. A SAFETY PRACTICE SHALL REQUIRE CONTRACTORS AT THEIR OWN DISCRETION.

5. CONTRACTOR SHALL SCHEDULE AND COORDINATE THE FOLLOWING WITH OWNER OR OWNER'S PROJECT MANAGER.
A. ANY AND ALL UTILITIES SHUT DOWN, CONTRACTOR SHALL GIVE A MINIMUM OF TWO WEEKS PRIOR NOTICE FOR EACH SHUT DOWN.
B. ANY AND ALL WORK BEING PERFORMED IN AREAS WHERE THE FACILITY WILL REMAIN IN OPERATION.
C. ANY WORK WITHIN THE PARAMETER OF ITEMS A & B ABOVE SHALL BE COORDINATED WITH THE GENERAL CONTRACTOR.

6. ALL MATERIALS AND WORKMANSHIP ARE SUBJECT TO APPROVAL BY OWNER. IF DEFECTIVE PORTION OF THE WORK SHALL BE REPLACED BY THE CONTRACTOR AT HIS PART OF THIS CONTRACT AT NO ADDITIONAL COST TO THE OWNER.

7. ABSOLUTE ACCURACY OF DRAWINGS CANNOT BE GUARANTEED, WHILE EVERY EFFORT HAS BEEN MADE TO COORDINATE THE LOCATION OF THE EXISTING DUCTWORK AND PIPING ETC. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO COORDINATE THE EXACT REQUIREMENTS GOVERNED BY ACTUAL JOB CONDITIONS AT NO ADDITIONAL COST TO THE OWNER.

8. IN THE AREA OF NEW CONSTRUCTION WILL BE DONE A NUMBER OF EXISTING SERVICES. CONTRACTOR SHALL PROTECT AND MAINTAIN ALL ACTIVE LINES IN GOOD OPERATIONAL CONDITION.

9. PROTECT EXISTING BUILDING STRUCTURES AND ADJACENT FINISHED SURFACES FROM DAMAGE BY WORK UNDER THIS DIVISION TO MATCH ADJACENT WORK. PROTECTING AND REPAIRING SHALL BE PERFORMED BY WORKMEN SKILLED IN THE TRADES INVOLVED.

10. BALANCING CONTRACTOR SHALL BALANCE ALL EQUIPMENT SHOWING PRESSURE DIFFERENTIALS AND VACUUM QUANTITIES AND PRESSES OR VACUUM INDICATED. BALANCING CONTRACTOR SHALL BE NEIGH CERTIFIED.

11. ALL MECHANICALS INVOLVED IN PIPE AND DUCT DEMOLITION SHALL WEAR PROPER PERSONAL PROTECTIVE GEAR AND SHALL MAKE THE NECESSARY PROVISIONS TO PREVENT EXPOSURE TO HAZARDOUS MATERIALS.

12. RELIEF OR OPEN FLAME ARE NOT PERMITTED DURING THE DEMOLITION PROCESS OF ANY FLAMMABLE OR COMBUSTIBLE PIPING OR EXHAUST DUCT SYSTEMS.

13. A JAIL WELDING PERMIT / HOT WORK SHALL BE OBTAINED FROM OWNER PRIOR TO COMMENCING ANY WELDING ACTIVITY.

14. CONTRACTOR SHALL SUBMIT DUCT MATERIAL SUBMITTALS TO OWNER AND TO THE DESIGN ENGINEER FOR REVIEW AND APPROVAL, PRIOR TO PURCHASING AND INSTALLATION.

15. CONTRACTOR SHALL COORDINATE FINAL INSPECTION OF COMPLETED WORK WITH DESIGN ENGINEER AND OWNER OR OWNER'S REPRESENTATIVE, FOR FINAL APPROVAL AND ACCEPTANCE.

16. ANY NEW OR EXISTING DUCTWORK OFFSETS REQUIRED AS A RESULT OF EXISTING JOB CONDITIONS OR LACK OF COORDINATION WITH OTHER TRADES, SHALL BE PROVIDED AT NO ADDITIONAL COST TO OWNER AND SUBJECT TO OWNER'S REVIEW.

17. COORDINATE LOCATION OF EXISTING DUCTWORK AND NEW DUCTWORK OR PIPE RACKS WITH OTHER TRADES.

18. CONTRACTOR SHALL PROVIDE DUCTWORK AND TRANSITIONS EQUAL TO FREE AREA OF DUCTWORK AS SHOWN ON DRAWING. PROVIDE ANY ADDITIONAL DUCTWORK AS REQUIRED TO ADD NO ADDITIONAL COST TO OWNER AND SUBJECT TO OWNER'S REVIEW.

19. ALL DUCTWORK SHALL BE SUPPORTED ACCORDING TO SMACHA AND DEMONSTRATED BRACE PER SMACHA'S DESIGN RESTRAINT MANUAL TO MEET BUILDING CODE.

20. MAINTAIN A MINIMUM HEAD CLEARANCE OF 84" ABOVE FINISHED FLOOR UNDER SUSPENDED DUCTWORK.

21. VERIFY WITH ENVIRONMENTAL HEALTH AND SAFETY DEPARTMENT (EH&S) THAT EXISTING DUCTWORK CONVEYING HAZARDOUS PRODUCTION MATERIALS (HPM) HAS BEEN DECONTAMINATED AND CLEARED AS SUCH PRIOR TO CUTTING INTO OR DEMOLISHING. EH&S REPRESENTATIVE SHALL BE PRESENT WHEN CONTRACTOR TAKES INTO DUCTWORK OR FIRING.

ABBREVIATIONS

ABREV	DESCRIPTION
BTH	BOTH THERMAL INLET, BOTH THERMAL UNIT PER HOUR
CAL	CALIFORNIA BUILDING CODE
CE	CEILING OUTLET
CHW	CHILLED WATER SUPPLY
CM	CONDENSER MEDIUM
CNDWR	CONDENSER WATER RETURN
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EXISTING MECHANICAL SCHEDULE - SHOWN FOR REFERENCE ONLY

EXISTING CONDENSING UNIT / FAN COIL SCHEDULE																	
TAG	MAKE	TYPE	LOCATION	LOCATION SERVED	REFRIGERANT		COMPRESSOR		COIL		ELECTRICAL			DIMENSIONS WxDH	OPERATING WEIGHT (LBS)	REMARKS	
					TYPE	R	QTY	V	PH	TH	FLA	MCA	MCCP				
(D) B112-RCU-1001	LEBERT	FAN COIL	BUE04-CG05547	B112 METAL SHED BUILDING	B112 BPHM ROOM	R22	SINGLE	1	5	208	3	60	68	83.6	90	30-12"x30-12"x7H	-
(D) B112-RCU-1001	LEBERT	CONDENSING UNIT	C5083	B112 AJOADENT ROOF	B119 BPHM ROOM	R22	N/A	1	5	208	3	60	-	-	-	-	-
(D) B132-RCU-1001	LEBERT	FAN COIL	BUE04-CG05547	B132 METAL SHED BUILDING	B132 BPHM ROOM	R22	SINGLE	1	5	208	3	60	68	83.6	90	30-12"x30-12"x7H	870
(D) B132-RCU-1001	LEBERT	CONDENSING UNIT	C5083	B132 AJOADENT ROOF	B132 BPHM ROOM	R22	N/A	1	5	208	3	60	-	-	-	-	-

* Connect refrigerant piping from condenser to respective fan coil. Field verify for exact routing, install per manufacturer's installation requirements, clearances, and refrigerant line limits (300 ft, consult factory if needed).
* Provide disconnect switch, #4 all equipment cords, LEBERT PCBs network controller, and Modine cooled coil.
* Complete disconnect piping to respective outdoor condenser. Field verify for exact routing, install per manufacturer's installation requirements, clearances, and refrigerant line limits (300 ft, consult factory if needed).

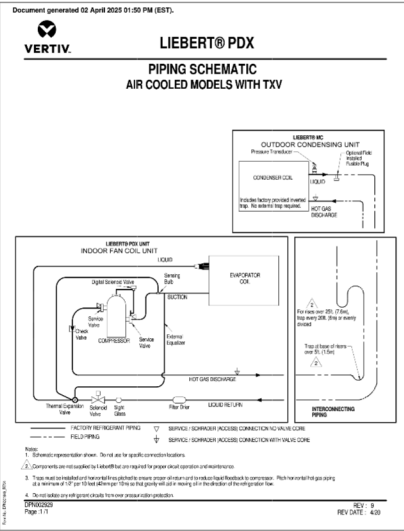
NEW MECHANICAL SCHEDULE

NEW AIR DISTRIBUTION SCHEDULE					
SYMBOL	MFR	MODEL	CFM	DESCRIPTION	REMARKS
CSD	TTUS	MCD	1000/1200	24X24 CEILING SUPPLY AIR DIFFUSER, 24X24 SQUARE NECK.	1
SWR-1	TTUS	SSFL	1600	30X18 SIDE WALL RETURN AIR GRILLE.	-
SWR-2	TTUS	SSFL	3200	30X30 SIDE WALL RETURN AIR GRILLE.	-

REMARKS:
 PROVIDE RAPID MOUNT FRAME AND OPPOSED BLADE DAMPER.

NEW CONDENSING UNIT / FAN COIL SCHEDULE																				
TAQ	MAKE	TYPE	MODEL	LOCATION	LOCATION SERVED	REFRIGERANT TYPE	REFRIGERANT CHARGE (LBS)	REFRIGERANT CONNECTION SIZE	COMPRESSOR	TONS	CFM	ESP (IN WC)	TSP (IN WC)	CLG CAP TOT MBH	SENS CLG CAP TOT MBH	ELECTRICAL	DIMENSIONS INxHxI	OPERATING WEIGHT (LBS)	REMARKS	
(N) B11-FCU-1001	VERTIV/LEBERT	FAN COIL	PX029	B118 METAL SHED BUILDING	B118 BPH ROOM	R454B	0.9	3/4" 1-1/8"	SINGLE	1	3200	0.28	0.75	155.3	-	-	208 3 160	34-1/2"x34-3/8"x77-1/2"	700	3.4, 5.7, 8
(N) B11-MRCU-1001	VERTIV/LEBERT	CONDENSING UNIT	PL059	B118 PARKING LOT	B118 BPH ROOM	R454B	1	3/4" 1-1/8"	SINGLE	1	8	-	-	-	-	-	208 3 60	88"x55-1/2"x35-7/8"	387	1.2, 4
(N) B132-FCU-1001	VERTIV/LEBERT	FAN COIL	PX029	B132 METAL SHED BUILDING	B132 BPH ROOM	R454B	0.9	3/4" 1-1/8"	SINGLE	1	3200	0.28	0.75	155.3	-	-	208 3 160	34-1/2"x34-3/8"x77-1/2"	700	3.4, 5.7, 8
(N) B132-MRCU-1001	VERTIV/LEBERT	CONDENSING UNIT	PL059	B132 PARKING LOT	B132 BPH ROOM	R454B	1	3/4" 1-1/8"	SINGLE	1	8	-	-	-	-	-	208 3 60	88"x55-1/2"x35-7/8"	387	1.2, 4
Remarks:																				
1. Connect refrigerant piping from condenser to respective fan coil. Field verify for exact routing. Install per manufacturer's installation requirements, clearances, and refrigerant line limits (300 ft. consult factory if needed).																				
2. Provide disconnect switch, 10/2700V ⁰ F ⁰ F ⁰ equipment code, Label POU network connection, and Modine isolated coil.																				
3. Connect refrigerant piping to respective outdoor condenser. Field verify for exact routing. Install per manufacturer's installation requirements, clearances, and refrigerant line limits (300 ft. consult factory if needed).																				
4. Provide 2-stage Electric Run Switch.																				
5. Refrigerant piping by manufacturer.																				
6. Provide disposable 34oz/2.1 MERV 11 filter.																				
7. Provide factory dual smoke detector for local automatic shut-down.																				

REFRIGERANT PIPING DIAGRAM (FOR REFERENCE ONLY)



REFRIGERANT CALCULATIONS									
CONDENSING UNIT	FAN COIL	ROOM	AREA (FT²)*	ESTIMATED HEIGHT (FT)	VOLUME (FT³)*	ASHRAE 15 R454B ALLOWABLE REFRIGERANT CHARGE LB/FT³*	ESTIMATED SYSTEM REFRIGERANT CHARGE (LB)	ALLOWABLE REFRIGERANT CHARGE (LB)	COMPLETS?
(N) 0-116-RCU-1001	(N) 0-116-FCU-1001	B116 META. SHED BUILDING	597	18.0	10748.0	3.1	6.9	33.3	YES
(N) 0-132-RCU-1001	(N) 0-132-FCU-1001	B132 META. SHED BUILDING	525	18.0	9450.0	3.1	6.9	29.3	YES

SAFETY NOTES		
1.	OPERATION OF ENERGY ISOLATING DEVICES SHALL BE PERFORMED BY SLAC WORKERS. CONTACT THE FIELD CONSTRUCTION MANAGER FOR ASSISTANCE WITH EQUIPMENT OPERATION.	
2.	SYSTEMS POTENTIALLY CONTAINING HAZARDOUS ENERGY SHALL BE LOCKED OUT IN ACCORDANCE WITH THE SLAC ESH MANUAL, CHAPTER 51.	
MECHANICAL SCOPE SUMMARY		
1.	DEMO B116 & B132 EXISTING LIBERTY FAN COIL AND OUTDOOR CONDENSER UNITS.	
2.	PARTIAL DEMO B116 SUPPLY AIR DUCT AND FULL DEMO OF B116 SUPPLY AND RETURN AIR DUCTS.	
3.	INSTALL NEW VERTIV FAN COIL IN B116 & B132 ELECTRICAL ROOM.	
4.	INSTALL NEW DUCTWORK, CEILING SUPPLY GRILLES, AND RETURN GRILLES IN B116 & B132.	
REFERENCE SYMBOLS		
SYMBOL	DESCRIPTION	
	DETAIL # SHEET #	
	SECTION # SCHEDULE	
	EQUIPMENT TAG # WINDOW TAG #	
	FEEDER TAG #	
	REVISION #	
	KEY NOTE #	
	POINT OF CONNECTION TO EXISTING (POC)	
	ELEVATION DATUM	
	EXISTING TO BE REMOVED (DEMOLISHED)	
	EXISTING TO REMAIN	
	NEW	
SYMBOLS AND ABBREVIATIONS LISTED ARE FOR GENERAL USE. DISREGARD THOSE WHICH ARE NOT USED ON THE DRAWINGS.		
DUCT INSULATION SCHEDULE TABLE PER CALIFORNIA TITLE 24 SECTION 120.4		
MANUFACTURER	TYPE	MINIMUM R-VALUE
REFLECTIX OR ECOLON	BUBBLE DUCT WRAP	R-8

DUCT SYMBOLS		
SINGLE LINE	DOUBLE LINE	DESCRIPTION
		SUPPLY AIR DUCT UP OR SECTION
		RETURN AIR DUCT UP OR SECTION
		EXHAUST AIR DUCT UP OR SECTION
		SUPPLY AIR DUCT DOWN
		RETURN AIR DUCT DOWN
		EXHAUST AIR DUCT DOWN
		CEILING SUPPLY DIFFUSER OR REGISTER
		CEILING RETURN AIR GRILLE
		CEILING EXHAUST AIR GRILLE
		WALL EXHAUST AIR GRILLE
		WALL RETURN AIR GRILLE
		WALL SUPPLY AIR DIFFUSER OR GRILLE
		FAN POWDERED HEPA FILTER
		WATER MOUNTED NONDUSTY LEPA HEPA FILTER
		WATER MOUNTED NONDUSTY SUPPLY SEC. HEPA FILTER
		W/FASE VELOCITY SUPPLYING SEC. HEPA F.
		DUCT WITH A BEND
		DUCT RISE OR DROP IN DIRECTION OF AIRFLOW
		1/2" (63.5) MAXIMUM SIZE NET INSIDE DIMENSIONS
		FLOODED DUCT
		RECTANGULAR TO ROUND TRANSITION
		FLEXIBLE CONNECTION
		90° (EL) W/FLEXING VANES (RIGID EL IS ACCEPTABLE)
		CONICAL 90° TEE
		90° TEE
		45° TEE
		RECTANGULAR TEE
		Y ~ FITTING
		45° HEAL FITTING RECTANGULAR
		ROUND OR RECTANGULAR REDUCER
		SPIN N FITTING WITH 45° EXTRACTOR IN RECTANGULAR MAIN
		AIR FLOW DIRECTION ARROWS

DAMPER LEGEND	
	BLIND/DAMPER (000)
	VARIABLE DAMPER (005)
	FIRE DAMPER (10)
	FIRE/DAMPER DAMPER (150)
	CONTROL DAMPER (20)
	CONTROL FIRE/DAMPER DAMPER (2-75)
	PARALLEL BLADES DAMPER (080)
	BACHMAN DAMPER (000)
	OSGOOD BLACK DAMPER (080)
	POSITIVE SEAL BUTTERFLY DAMPER W/MOTOR (085)
	MOTORIZED OSGOOD BLACK DAMPER
	MOTORIZED PARALLEL BLADES DAMPER

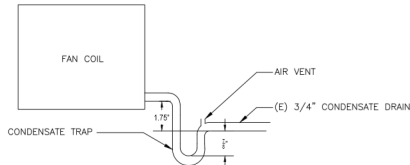
SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (FIC)	
BY: ondlar	
DATE: 1/23/2026	PRID: 25-254
Revised Authorization	☒ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and ESR program requirements. The authorization of these plans shall not be construed to be a permit or approval for violation of any applicable codes, standards, or ESR programs.	
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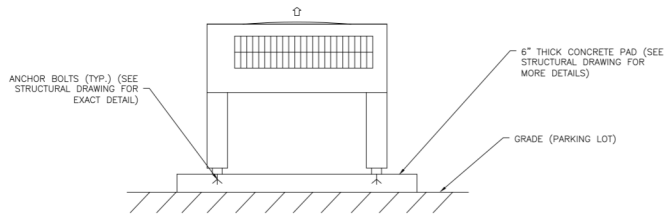


SCALE AS NOTED		DO NOT SCALE DRAWING	55010.2025.01	M.0.01	
 			B116 AND B132 B116 & B132 HVAC DESIGN MECHANICAL COVER SHEET		
THE DRAWINGS, SPECIFICATIONS AND DATA HEREIN PROVIDED SHALL NOT BE USED FOR ANY PURPOSES OTHER THAN OBSERVATION WITHOUT THE WRITTEN PERMISSION OF STAFF AND UNIVERSITY, SLAC					
EDR# 0000-00000000 DESIGNED BY CHECKED BY DATE	DATE 01/10/2025 EDR#0000-00000000	DATE 01/10/2025	DRAWING NUMBER B0116-M00-00002	REVISION NUMBER D	E

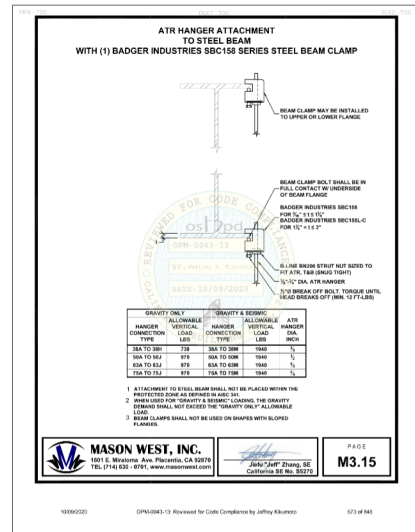
REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	KT	JL	MHC	10-02-2025
B	100% CD	KT	JL	MHC	10-27-2025
C	FAO REVIEW	KT	JL	MHC	11-21-2025
D	BIO REVIEW	KT	JL	MHC	12-16-2025



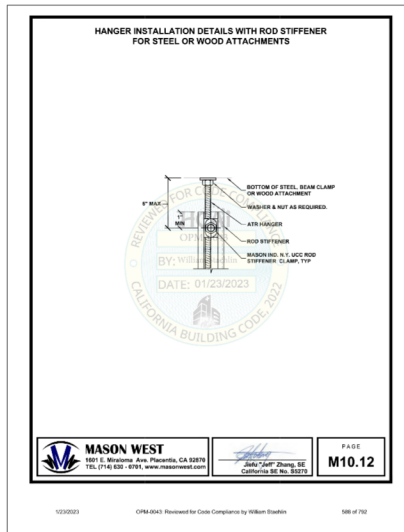
① CONDENSATE DRAIN TRAP DETAIL
SCALE: N.T.S.



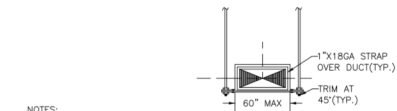
② FLOOR MOUNTED CONDENSER DETAIL
SCALE: N.T.S.



③ STEEL BEAM CLAMP DETAIL
SCALE: N.T.S.



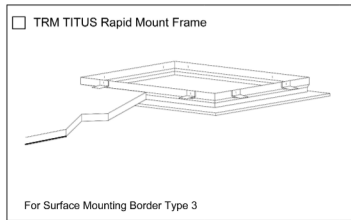
④ ROD STIFFENER ATTACHMENT DETAIL
SCALE: N.T.S.



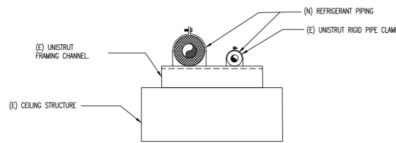
- NOTES:
- HANGER SPACING WILL BE 6'.
 - SWAY & SEISMIC BRACING IS NOT REQUIRED PER ASCE 7-16 EXCEPTION LISTED BELOW.
2.1.1. THE DUCT HAS A CROSS-SECTIONAL AREA LESS THAN 6.0 SQUARE FEET.
2.1.2. THE DUCT WEIGHS 20 POUNDS PER FOOT OR LESS.
 - HANGER MATERIAL SUPPORTING FLEXIBLE DUCT SHALL IN NO CASE BE LESS THAN 1/2" WIDE. FLEXIBLE DUCT SHALL BE SUPPORTED PER MANUFACTURER RECOMMENDED MATERIALS, BUT AT NO GREATER DISTANCE THAN 4FT MAX. PERMISSIBLE SAG IS MAX. 1" PER FOOT OF SPACING BETWEEN SUPPORTS.

RECTANGULAR DUCT HANGERS MINIMUM SIZE				
MAX DUCT LOAD (POUNDS)	DUCT SPACING (FT)	MIN DUCT SPACING (IN)	MIN DUCT SPACING (IN)	MIN DUCT SPACING (IN)
1/2" - 1"	1" x 1/2" IN	1" x 1/2" IN	1" x 1/2" IN	1" x 1/2" IN
1" - 1 1/2"	1" x 3/4" IN	1" x 3/4" IN	1" x 3/4" IN	1" x 3/4" IN
1 1/2" - 2"	1" x 1" IN	1" x 1" IN	1" x 1" IN	1" x 1" IN
2" - 4"	1" x 1 1/2" IN	1" x 1 1/2" IN	1" x 1 1/2" IN	1" x 1 1/2" IN

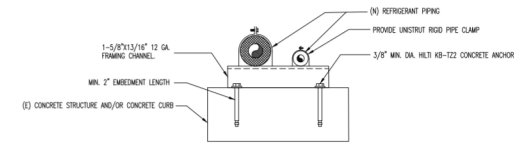
⑤ TYPICAL DUCT HANGER DETAIL
SCALE: N.T.S.



⑥ EXISTING HARD CEILING DIFFUSER MOUNTING FRAME DETAIL
SCALE: N.T.S.



⑦ EXISTING REFRIGERANT PIPE SUPPORT DETAIL
SCALE: N.T.S.



⑧ NEW EXTERIOR REFRIGERANT PIPE SUPPORT DETAIL
SCALE: N.T.S.

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (FC)	
BY: <u>order</u>	DATE: 10/20/2025
PROJECT: 25-204	REVISION: 01
These plans have been approved for construction by the Building Inspection Office of the State of California. The Building Inspection Office is not responsible for the design or construction of the project.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	



SCALE: AS NOTED		DO NOT SCALE DRAWING		S5010 2025.01		M0.02	
ENGR: <u>CHEN/SHU</u>		DATE: 10/20/25		APPROVALS: <u>SHU</u>		DATE: 10/20/25	
DES: <u>WAL/CHEN</u>		CHK: <u>WAL</u>		REV: <u>01</u>		REV: <u>01</u>	
DRAWING NUMBER: B0116-M05-0003		REVISION: D		DATE: 10/20/25		DATE: 10/20/25	

B116 AND B132
B116 AND B132 HVAC Design
MECHANICAL
DETAILS

B0116-M05-0003 D E

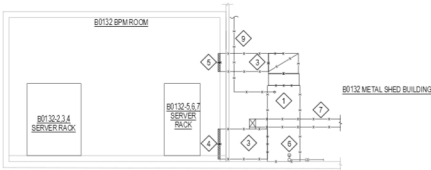
REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	KT	JL	MHC	10-02-2025
B	100% CD	KT	JL	MHC	10-27-2025
C	FAO REVIEW	KT	JL	MHC	11-21-2025
D	BIO REVIEW	KT	JL	MHC	12-16-2025

PLAN GENERAL NOTES

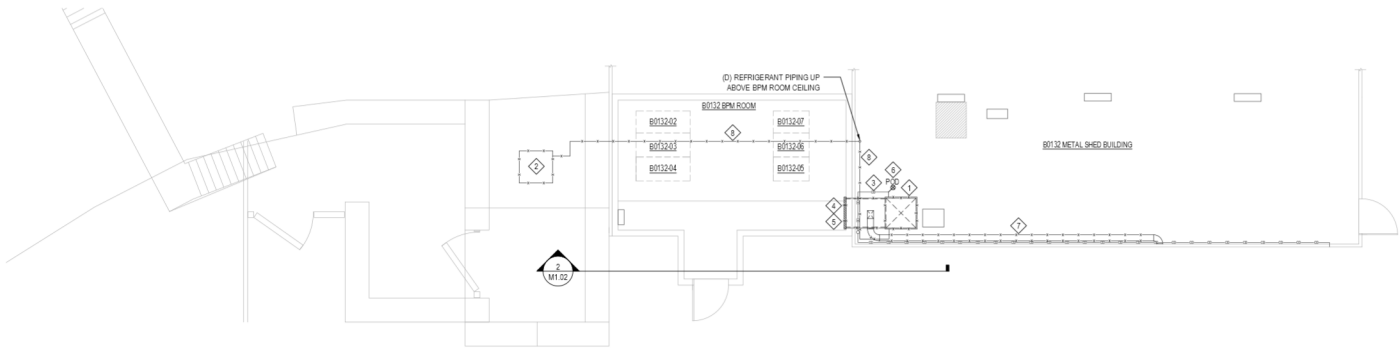
- EXISTING MECHANICAL TO REMAIN, UNLESS OTHERWISE NOTED.
- SEE ELECTRICAL DRAWINGS FOR ALL ELECTRICAL WORK.

MECHANICAL SHEET NOTES

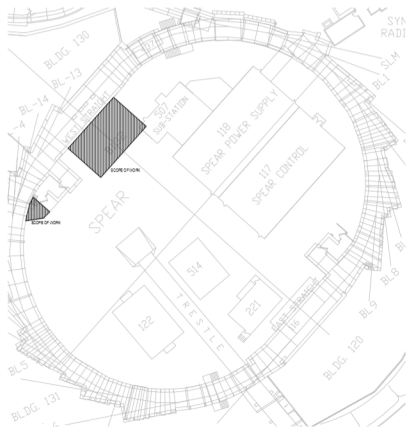
- DEMO EXISTING 5-TON INDOOR LIEBERT FANCOIL.
- DEMO EXISTING 5-TON OUTDOOR LIEBERT CONDENSING UNIT ON ROOF.
- DEMO EXISTING SUPPLY DUCTWORK ABOVE AND RETURN DUCTWORK BELOW.
- DEMO EXISTING FLOOR LEVEL 30X30 SIDE RETURN AIR GRILLE. PATCH AND FILL WALL WITH MATERIAL MATCHING EXISTING CONDITIONS.
- DEMO EXISTING CEILING LEVEL 30X18 SIDE WALL SUPPLY AIR GRILLE.
- PARTIAL DEMO EXISTING 3/4" CONDENSATE DRAIN FROM EXISTING FAN COIL AS SHOWN ON PLAN.
- DEMO EXISTING 8X8 OUTSIDE AIR DUCTWORK. PATCH AND FILL WALL PENETRATION TO EXTERIOR.
- DEMO EXISTING REFRIGERANT PIPING BETWEEN EXISTING FAN COIL AND OUTDOOR CONDENSING UNIT.



2 B0132 - MECHANICAL SECTION DEMO PLAN
SCALE: 1/4" = 1'-0"



1 B0132 - MECHANICAL DEMO PLAN
SCALE: 1/4" = 1'-0"



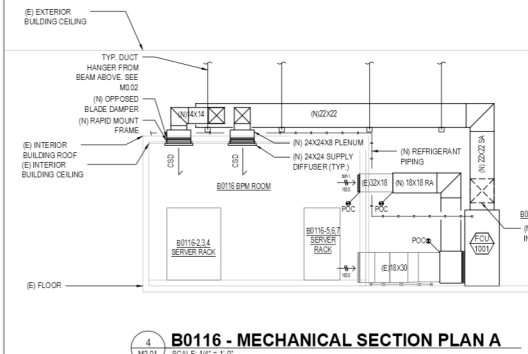
3 SITE PLAN
SCALE: NTS

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

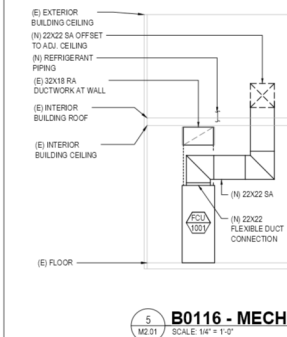
SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (PC)	
BY: ☐ ☒	PRM: ☐ ☒
DATE: 10/20/2025	PRM: 25-204
Revised Authorization ☒ No ☐ Yes	
These plans have been inspected and approved for construction. No further changes are allowed without the written approval of the SLAC Building Inspection Office.	
The authorization of these plans does not constitute a warranty or endorsement of any products, materials, or methods.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	



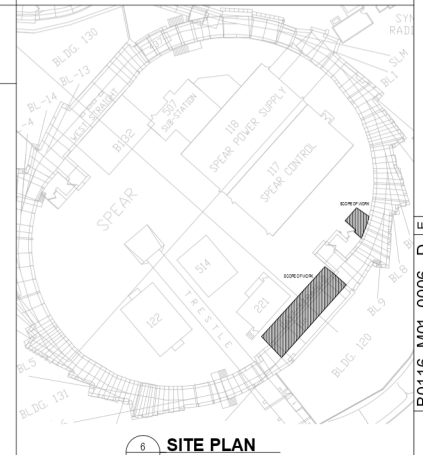
SCALE: AS NOTED	DO NOT SCALE DRAWING	5/5010 2025.01	M1.02
SLAC ENERGY		B132 B116 AND B132 HVAC Design MECHANICAL DEMO PLAN	
THE DRAWINGS, SPECIFICATIONS AND OTHER DATA HEREIN PROVIDED SHALL NOT BE COPIED, PUBLISHED OR OTHERWISE FURTHER DISSEMINATED WITHOUT PRIOR WRITTEN PERMISSION OF SLAC.		DRAWING NUMBER B0116-MD1-0005	
ENGR: ☐ ☒	DATE: ☐ ☒	APPROVALS: ☐ ☒	REVISION: ☐ ☒
DES: ☐ ☒	DATE: ☐ ☒	APPROVALS: ☐ ☒	REVISION: ☐ ☒
CHK: ☐ ☒	DATE: ☐ ☒	APPROVALS: ☐ ☒	REVISION: ☐ ☒



4 B0116 - MECHANICAL SECTION PLAN A
M2.01 SCALE: 1/4" = 1'-0"



5 B0116 - MECHANICAL SECTION PLAN B
M2.01 SCALE: 1/4" = 1'-0"



6 SITE PLAN
M2.01 SCALE: NTS

SCALE: AS NOTED	DO NOT SCALE DRAWING	S5010 2025.0
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MHC ENGINEERS
150 8TH STREET
SAN FRANCISCO, CA 94103
PH. (415) 512-7141
FAX (415) 512-7120



 SLAC SATURN ACCELERATOR LABORATORY	 U.S. DEPARTMENT OF ENERGY	THE DRAWINGS, SPECIFICATIONS AND OTHER DATA HEREIN PROVIDED SHALL NOT BE COPIED, REPRODUCED OR OTHERWISE FURTHER DISSEMINATED WITHOUT PRIOR WRITTEN PERMISSION OF SLAC/ ORNL UNIVERSITY / SLAC.	
		ENGR CHEN, HENG-HUO DES TAN, KUN CALZ	DATE 10-10-2021 APPROVALS (SIGNED AND) DATE 10-10-21

B116
B116 AND B132 HVAC Design
MECHANICAL
PLAN

DRAWING NUMBER	REVISION	
B0116-M01-0006	D	E

M2 01

B0132 - MECHANICAL ABOVE CEILING PLAN

2 B0132 - MECHANICAL RCP PLAN

4
M2.02

B0132 - MECHANICAL SECTION PLAN A

SCALE: 1/4" = 1'-0"

5 B0132 - MECHANICAL SECTION PLAN B
M2.02 SCALE: 1/4" = 1'-0"

B0132 - MECHANICAL PARKING LOT AND FLOOR PLAN

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: snadler	
DATE: 1/23/2026	FRSE: 25-254
Revised Authorization	☒ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and IBC program requirements. The authorization of these plans shall not be construed to be a permit or approval for violation of any ASD/ACCM codes, standards, or EIR programs.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	



SCALE: AS NOTED	DO NOT SCALE DRAWING	S5010 2025.0
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B132		MECHANICAL	
B116 AND B132 HVAC Design			
MECHANICAL			
PLAN			
DRAWING NUMBER		REVISION	
B0116-M01-0007		D	E

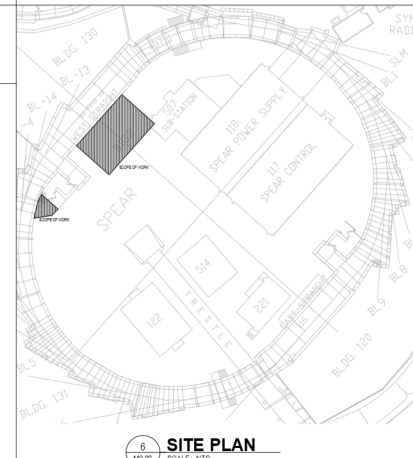
REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	KT	JL	MHC	10-02-2025
B	100% CD	KT	JL	MHC	10-27-2025
C	FAO REVIEW	KT	JL	MHC	11-21-2025
D	BIO REVIEW	KT	JL	MHC	12-18-2025

PLAN GENERAL NOTES

- 1 EXISTING MECHANICAL REMAIN, UNLESS OTHERWISE NOTED.
- 2 SEE ELECTRICAL DRAWINGS FOR ALL ELECTRICAL WORK.
- 3 PROVIDE MANUAL VOLUME BALANCING DAMPERS AT EACH BRANCH TAKE-OFF IN ALL RETURN DUCTWORK.
LOCATE BALANCING DAMPERS AS CLOSE TO THE BRANCH TAKE-OFF AS POSSIBLE.
- 4 CONTRACTOR SHALL PROVIDE ALL MATERIALS, SUPPLIES, SUBMIT TO OWNER AND TO THE DESIGN ENGINEER FOR REVIEW AND APPROVAL, PRIOR TO PURCHASING AND INSTALLATION.
- 5 ANY NEW OR EXISTING DUCTWORK OFSETS BE REQUIRED AS A RESULT OF EXISTING JOB CONDITIONS OR
DESIGN OF CONTRACTOR. ALL EXISTING TRACES, SHALL BE PROVIDED AT NO ADDITIONAL COST TO OWNER
AND SUBJECT TO OWNER'S REVIEW.
- 6 CONTRACTOR SHALL PROVIDE DUCTWORK AND TRANSITIONS EQUAL TO FREE AREA OF DUCTWORK AS
SHOWN ON DRAWING, TO PREVENT A CONFICT WITH EXISTING CONDITIONS.
- 7 ALL DUCTWORK SHALL BE SECURED TO STRUCTURE TO MATCHMA AND SECURELY BRACE PER MATCHMA'S
DESIGN RESTRAINT MANUAL TO MEET BUILDING CODE.
- 8 TAB CONTRACTOR SHALL BALANCE ALL EQUIPMENT SHOWN ON THIS DOCUMENT PER THE FOLLOW
QUANTITIES.
- 9 CONTRACTOR SHALL COORDINATE WITH FINAL INSPECTION OF COMPLETED WORK, WITH DESIGN
ENGINEER AND OWNER OR OWNER'S REPRESENTATIVE, FOR FINAL APPROVAL AND ACCEPTANCE.
- 10 CONTRACTOR SHALL VERIFY ALL EXISTING BUILDING CONDITIONS.

MECHANICAL SHEET NOTES

- ④ PROVIDE NEW LEIBERT P2200 1/2" IN. UPFLOW UNIT. INSTALL 30" - 2" FROM EXISTING LOCATION FROM THE DISCHARGE WALL THAN THE OLD FAN COIL. MANTAIN CODE REQUIRED CLEARANCES TO THE ELECTRICAL PANELS AND EQUIPMENT. PROVIDE DET SMOKE DETECTOR FOR LOCAL AUTOMATIC SHUTDOWN. FAN COIL SHALL BE LOCALLY CONTROLLED AND MAINTAIN 68°F (+1°). FIELD VERIFY FOR ANY OBSTRUCTION. PROVIDE FLOOR MOUNTING PER STRUCTURAL DRAWINGS.
- ④ PROVIDE NEW 36X12X24 RETURN AIR FLEXIBLE CONNECTION TO RETURN AIR INLET OF FANCOIL UNIT. COMPLETE WITH DETECT INSULATION.
- ④ PROVIDE NEW 36X12X24 CONDENSATE DRAIN LINE FROM FANCOIL AND EXTEND AND RE-CONNECT TO EXISTING 3/4"X3 CONDENSATE DRAIN. TERMINATE TO HUB DRAIN AS SHOWN ON PLAN. VERIFY FOR EXACT LOCATION.
- ④ PROVIDE NEW 36X18 RETURN AIR DUCT WITH 1" INSULATION.
- ④ PROVIDE NEW 24X24 SUPPLY AIR DUCT FROM FANCOIL OUTLET AND OFFSET TO ADJACENT ROOM CELING. PROVIDE FLEXIBLE DUCT CONNECTION AT FANCOIL OUTLET. PROVIDE DUCT INSULATION AND UNISTRUT DUCT SUPPORTS FROM CELING STRAP BEAMS. SEE DETAILS ON MO.02.
- ④ PROVIDE NEW 30X20 SIDEWALL RETURN AIR GRILLE AT EXISTING WALL OPENING CLOSE TO CEILING. ENLARGE OPENING FOR LARGER GRILLE SIZE.
- ④ EXISTING CEILING LOCATIONS TO REMAIN. COORDINATE WITH ELECTRICAL AND FIELD VERIFY FOR EXACT LOCATION.
- ④ PROVIDE NEW 4-WAY ADJUSTABLE 24X24 CELING FAN SUPPLY AIR DIFFUSER. COMPLETE WITH OPPOSED BLADE DAMPER AND RAPID MOUNT CELING FRAME.
- ④ PROVIDE NEW 24X24 FLEXIBLE ABOVE 24X24 CELING FAN SUPPLY AIR DIFFUSER AND CONNECT TO NEW 3/4"X18 BRANCH DUCT.
- ④ PROVIDE NEW 14X14 SUPPLY AIR BRANCH DUCT. PROVIDE DUCT INSULATION AND UNISTRUT DUCT SUPPORTS FROM CELING STRAP BEAMS. SEE DETAILS ON MO.02.
- ④ PROVIDE NEW LEIBERT MCL06 OUTDOOR CONDENSING UNIT ON 8"X30"X6 INCRETE PAD. INSTALL PER MANUFACTURERS INSTALLATION REQUIREMENTS AND CLEARANCES. EQUIPMENT IS LOCATED IN A SECURE LOCATION. COORDINATE WITH ELECTRICAL FOR POWER CONNECTION. REFER TO STRUCTURAL DRAWINGS FOR MOUNTING DETAILS.
- ④ NEW REFRIGERANT PIPING BETWEEN NEW FANCOIL AND OUTDOOR CONDENSING UNIT. USE EXISTING UNISTRUT MOUNT. PROVIDE NEW UNISTRUT MOUNT ABOVE EXISTING CONCRETE W/RT AS REQUIRED. REFRIGERANT PIPING EXPOSED TO WEATHER SHALL BE PROVIDED WITH A WATER RETARDANT COVER OVER THE INSULATION. ACHIEVE THIS NOT YET USED FOR PROTECTION. FIELD VERIFY FOR EXACT EXISTING UNISTRUT LOCATIONS, ROUTING, AND CONDITIONS.
- ④ NEW 9" Ø BOLLARD. COORDINATE WITH SLAC TO CONFIRM EXACT NUMBER OF BOLLARDS AND LOCATION.
- ④ NEW UNISTRUT DUCT SUPPORT EVERY 6' FROM STEEL BEAM ABOVE. PROVIDE BEAM CLAMP 3/8" Ø THREADED ROD, ROD STOPPER, AND UNISTRUT SUPPORTS. SEE DETAILS ON MO.02. FIELD VERIFY.



6 **SITE PLAN**
M2.02 SCALE: NTS

B0116- M01 - 0007	D	E
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REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	HF	JO	SK	10-06-2025
B	100% CD	HF	JO	SK	11-06-2025
C	FAO REVIEW	HF	JO	SK	11-24-2025
D	BIO REVIEW	HF	JO	SK	12-18-2025

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"Where engineering and technology drive innovation"
 2603 Camino Ramon, Suite 417
 San Ramon, CA 94583
 +1 (866) VEKTOR1 (835-8671)



B0116 - E00 - 0008 - D	E
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(N) ELECTRICAL EQUIPMENT SCHEDULE											
TAG	(APPROXIMATE) QUANTITY	DESCRIPTION	MANUFACTURER CATALOG NUMBER	VOLTAGE (V)-(L)-[A]-[N]	PHASE	FRAME RATING (AMPS)	US CONDUCTOR RANGE, PER PHASE (AWG)	DIMENSIONS (EA, INT.)	MOUNTING TYPE	APPROXIMATE WEIGHT (LBS.)	REMARKS & AREA OF USE
	2	HEAVY DUTY SAFETY SWITCH, NON FUSIBLE, 30A, 3-PM, RATED 600VAC, UL LISTED, OUTDOOR RAIN PROOF, UL COULDSIDE COMPLY W/ OSHA REQUIRED HARDWARE AND GROUND KIT, UL LISTED, (1)-YEAR WARRANTY.	SEWING INFABUS 30A (3-PM) SWITCH H02324 (GROUND KIT)	208V/-	3	30A	#4-6AWG	11" x 7" x 8" x 7" 10.17"	SURFACE	11.05	NO CONDUITOR DISCONNECT SEE 100-1000 AND 8131-205-1001
NOTES:											
1. INSTALL (N) EQUIPMENT LISTED IN THIS SCHEDULE COMPLETE, W/ALL REQUIRED HARDWARE.											
2. INSTALL (N) EQUIPMENT PER MANUFACTURER'S INSTRUCTIONS (UNLESS CHANGES NOTED)											
3. STRUCTURAL CALCULATIONS ARE NOT REQUIRED PER ASCE 7-18 CHAPTER 13, AND PER S&C SEISMIC DESIGN SPECIFICATION FOR BUILDINGS, STRUCTURES, EQUIPMENT, AND SYSTEMS - 2003 EDITION - SECTION 1.3.											

<u>(N) GND. IN-GRADE ENC. SCHEDULE:</u>								
TAG	(APPROXIMATE) QTY	MANUFACTURER	MANUFACTURER CATALOG NUMBER	MIN. EXTERIOR DIMENSIONS (EA, UNIT) DIA. X D"	MIN. INTERIOR DIMENSIONS (EA, UNIT) DIA. X D"	LD TYPE	ETCHED MARKING	REMARKS & AREA OF USE
PS01	2	QUICKCARE INFRASTRUCTURE	F08	12-1/2" DIA. X 12"	8" DIA. X 8"	FIBER	"Q01"	GROUND ROD ENCLOSURE

NOTES:

1. PROVIDE ALL REQUIRED HARDWARE AND EXPANSION RINGS, COVER SHALL BE ETCHED "Q01" W/ FASTENERS.

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (FC)	
BY: <u>ondier</u>	
DATE: <u>1/23/2026</u>	PR#: <u> 25-254</u>
Revised Authorization ☒ No ☐ Yes	
These plans have been reviewed for compliance with applicable building codes and standards, and EIR program requirements.	
The authorization of these plans shall not be construed to be a permit or an approval for violation of any applicable codes, standards, or EIR programs.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

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"Where engineering and technology drive innovation"
 2603 Camino Ramon, Suite 417
 San Ramon, CA 94583
 +1 (866) VEKTOR1 (835-8671)



SCALE AS NOTED		DO NOT SCALE DRAWING	55010 2025.01	E0.02	
 			B116 AND B132 B116 AND B132 HVAC DESIGN ELECTRICAL EQUIPMENT SCHEDULES		
THE DRAWINGS, SPECIFICATIONS AND OTHER DATA HEREIN PROVIDED SHALL NOT BE REPRODUCED OR TRANSMITTED IN ANY FORM OR BY ANY MEANS WITHOUT THE WRITTEN PERMISSION OF SLAC AND/OR THE U.S. DEPARTMENT OF ENERGY.					
ERROR: (CHECK) (BIBI): DATE: 10/20/2024 CDR: [Signature]	DATE: 10/20/2024 APPROVALS: [Signature] DATE: 10/20/2024	DRAWING NUMBER: B0116-E06-0009		REVISION: D	E

DEMOLITION GENERAL NOTES

- [illegible]

DEMOLITION KEY NOTES

- 1 SEE MECHANICAL SHEETS FOR REMOVAL OF (E) EQUIPMENT.
- 2 REMOVE (E) FEEDER BETWEEN (E) FAN COIL AND (E) DISCONNECT SWITCH.
- 3 DISCONNECT (E) CONDUIT(S) AT (E) CONDENSER DISCONNECT SWITCH AND REMOVE W/ FLEX TO NEAREST UNION. PULL OUT (E) CONDUIT(S) TO SOURCE PANEL. REMOVE (E) DISCONNECT SWITCH. PREPARE (E) CONDUIT SECTION FOR RE-USE.
- 4 DISCONNECT (E) HVAC CONTROL CONDUIT(S) AND REMOVE W/ FLEX TO THE NEAREST UNION. REMOVE ALL (E) CONDUITS BETWEEN (E) CONDENSER AND (E) FANCOIL. PREPARE (E) CONDUIT SECTION FOR RE-USE.
- 5 DISCONNECT (E) FEEDER OUT(S) FROM (E) PANEL. REMOVE CONDUIT TO THE NEAREST UNION. REMOVE (E) CONDUIT(S) BETWEEN (E) PANEL TO (E) CONDENSER DISCONNECT SWITCH.



All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.



SCALE: AS NOTED		DO NOT SCALE DRAWING	55010 2025 01		E3.01D
 			B116 And B132 B116 And B132 HVAC DESIGN ELECTRICAL DEMO PLAN		
THE DRAWINGS, SPECIFICATIONS AND OTHER DATA HEREIN PROVIDED SHALL NOT BE USED, IN WHOLE OR IN PART, FOR OTHER PROJECTS OR REPRODUCTION WITHOUT THE WRITTEN PERMISSION OF SLAC AND/OR DOE.					
ENDR. NO. 1 DATE: 08/01/00 DESIGNED BY: JLM CHECKED BY: JLM	DATE: 01/01/01 01/02/01 01/03/01	APPROVALS DATE: 01/03/01	DRAWING NUMBER B0116-ED1-0001		REVISION D E

REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	HF	JO	SK	10-06-2025
B	100% CD	HF	JO	SK	11-06-2025
C	FAO REVIEW	HF	JO	SK	11-24-2025
D	BIO REVIEW	HF	JO	SK	12-18-2025

PLAN GENERAL NOTES

- INFORMATION PROVIDED ON THESE CONSTRUCTION DOCUMENTS WAS GATHERED FROM FIELD OBSERVATION AND PREVIOUS RECORD DRAWINGS AND SHALL BE USED WITH CAUTION. CONTRACTOR SHALL FIELD VERIFY ALL QUANTITIES, ESPECIALLY THE EXACT COUNT OF DEVICES, ROUTING, SIZES, LENGTHS AND COORDS OF CONDUIT AND CONDUITORS TO BE INSTALLED AND OR REMOVED PRIOR TO STARTING THE RESPECTIVE WORK.
- FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS SEE SHEET E01.
- WORK SHALL NOT INTERFERE WITH ADJACENT FACILITY USE AND OR FUNCTION. CONTRACTOR SHALL COORDINATE WITH OWNER AT LEAST 2-WEEKS PRIOR TO STARTING WORK TO ESTABLISH A MUTUALLY AGREED UPON SCHEDULE.
- CONTRACTOR SHALL COORDINATE ALL WORK WITH ALL OTHER TRAVEL.
- CONTRACTOR SHALL MAINTAIN AND VERIFY CONTINUITY OF UPSTREAM AND DOWNSTREAM ELECTRICAL CIRCUITS DURING AND AFTER WORK IS COMPLETE.
- CONTRACTOR SHALL FOLLOW MANUFACTURER'S MOUNTING, WIRING AND INSTALLATION INSTRUCTIONS FOR ALL ASSEMBLIES AND DEVICES UNLESS OTHERWISE NOTED.
- SEE EQUIPMENT SCHEDULE ON SH. E03.02 FOR RESPECTIVE EQUIPMENT TAGS.
- SEE FEEDER SCHEDULE ON SH. E03.02 FOR RESPECTIVE FEEDER TAGS.

PLAN KEY NOTES

- SEE MECHANICAL SHEETS FOR PROVISION OF (N) EQUIPMENT AND CONC. PAD DIMENSIONS.
- PROVIDE (N) 200VAC 3PH 30A DISCONNECT SWITCH. PROVIDE WP FLEX FROM (N) DISCONNECT SWITCH TO (N) CONDENSER HIGH VOLTAGE ENTRANCE.
- PROVIDE (N) IN-USE WP GFCI NEMA 5-20R DUPLEX RECEPTACLE.
- PROVIDE (N) FEEDER USING 1-1/2" MC W/BRANDS + JGCS FROM (E) DISC. SWITCH TO (N) FAN FOR HIGH VOLTAGE ENTRANCE. LAST 3FT WP FLEX FOR EASE OF INSTALLATION.
- PROVIDE (N) FEEDER USING (E) CONDUIT ROUTING UP TO (N) J-BOX ON ROOF AND TRANSITION TO (N) 1" MC W/ SPACING + JGCS. TERMINATE AT (N) CONDENSER DISCONNECT SWITCH. LAST 3FT WP FLEX FOR EASE OF INSTALLATION.
- PROVIDE (N) POX NETWORK ROUTING USING (E) ROUTING FROM (N) FAN COIL TO (N) J-BOX ON ROOF AND TRANSITION TO (N) 1" MC W/AN NYLON ROPE. TERMINATE AT (N) CONDENSER LOW VOLTAGE ENTRANCE. LAST 3FT WP FLEX FOR EASE OF INSTALLATION.
- PROVIDE (N) 120VAC 20A BRANCH Ckt. USING 3/4" MC W/ 20'FANCOIL + 130' TO (N) DUPLEX RECEPTACLE.
- PROVIDE (N) 3/4" DIA X 10'-0" LONG CUL GROUND ROD IN (N) IN-SHADE ENCLOSURE. BOND (N) CONC. PAD REBAR AND (N) CONDENSER SUPPORT STRUCTURES TO (N) GROUND ROD. USE 3/4"MC W/BRANDS STRANDED BARE COPPER REC FOR INTERCONNECTION WPFC BFC AND TRANSITION TO GALEY MC W/FC PIPE WRAP ENTIRE TRANSITION ABOVE AND BELOW GRADE.
- PROVIDE (N) WP 10TH X 10TH X 8'0" J-BOX FOR TRANSITION FROM (E) CONDUIT ROUTING TO (N) CONDUIT ROUTING TO CONDENSER.
- PROVIDE (N) WP FLEX FROM NEAREST UNION TO (N) FAN COIL LOW VOLTAGE ENTRANCE.

1 B116 RF CTRL. ROOM HVAC

SCALE: 1/4" = 1'-0"



2 B132 RF CTRL. ROOM HVAC

SCALE: 1/4" = 1'-0"



SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (BFC)	
BY: trader	
DATE: 12/2/2025	PRJ#: 25-264
Revised Authorization ☒ No ☐ Yes	
This authorization is provided for the purpose of permitting the contractor to perform the work shown on these plans and specifications, and is not a guarantee of the accuracy or completeness of the information provided. The contractor shall be responsible for verifying the accuracy and completeness of the information provided.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

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 2603 Camino Ramon, Suite 417
 San Ramon, CA 94583
 +1 (925) Vektor1 (535-8871)



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SLAC U.S. DEPARTMENT OF ENERGY NATIONAL ACCELERATOR LABORATORY		B116 AND B132 B116 AND B132 HVAC DESIGN ELECTRICAL FLOOR PLAN	
ENGINEER: HENRI SHAPIRO DESIGNED BY: HENRI SHAPIRO CHECKED BY: HENRI SHAPIRO	DATE: 10-06-2025 APPROVALS: HENRI SHAPIRO DATE: 10-06-2025	DRAWING NUMBER: B0116-EP1-0012	REVISION: D

REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	HF	JO	SK	10-06-2025
B	100% CD	HF	JO	SK	11-06-2025
C	FAO REVIEW	HF	JO	SK	11-24-2025
D	BIO REVIEW	HF	JO	SK	12-18-2025

PANELS GENERAL NOTES

- CONTRACTOR SHALL COMPLY WITH ALL NOTES AS LISTED FOR EACH RESPECTIVE PNL.
- CONTRACTOR SHALL PROVIDE (N) TYPED PANEL SCHEDULE FOR EACH ELECTRICAL PANEL, PLACE IN CLEAR PLASTIC SHEET AND ATTACH TO INNER SURFACE OF PANEL DOOR.
- SUBCONTRACTORS SHALL NOT PERFORM WORK ON LEGACY SINGLE-PHASE CIRCUITS OR IN LEGACY ELECTRICAL PANELBOARDS (LEGACY = INSTALLED PRIOR TO 2010) THAT CARRY SINGLE PHASE CIRCUITS UNLESS NEUTRAL WIRING VERIFICATION OF ALL SINGLE-PHASE CIRCUITS IN THE PANEL HAS BEEN PERFORMED AND DOCUMENTED IN ACCORDANCE WITH SLAC-708-404-OR OPERATIONS OR SYSTEMS WITH SUSPECTED SHARED NEUTRALS. SEE ESY MANUAL, CHAPTER 8 ELECTRICAL SAFETY MANUAL, SECTION 7.2.3. IF THE VERIFICATION HAS NOT BEEN PERFORMED, THEN IT MUST BE PERFORMED BY EDP PNO PRIOR TO START OF PROJECT CONSTRUCTION.

PANELS KEY NOTES

- VERIFYING PANEL SCHEDULE VERIFIED BY AND SIGNED PNO ON 08/20/2025
- ALL ELECTRICAL PANELS SHALL BE INSPECTED FOR SHARED NEUTRALS BY ELECTRICAL SUBCONTRACTOR PRIOR TO BEING RE-ENERGIZED, REPORT FINDINGS TO EEDR AND OWNER REPRESENTATIVES PRIOR TO PROCEEDING.

	VOLTS:	208Y/120																																													
	PHASE:	3	4																																												
	WIRING:	250A																																													
	POLES:	42																																													
	LOAD DESCRIPTION	TYPE	A	B	C	BRK	CKT	BRK	A	B	C	TYPE	LOAD DESCRIPTION																																		
	RING - LIGHTING	L	0.30			201	1	2	201	0.54	0.54	R	RING - RECEPT																																		
	RING - LIGHTING	L	0.30			201	3	4	201	0.54	0.54	R	RING - RECEPT																																		
	RING - LIGHTING	L	0.30			201	5	6	201	0.54	0.54	R	RING - RECEPT																																		
	SPARE	-	0.00			201	9	10	300	1.20	1.20	R	RECEPT - 30A																																		
	EM. LIGHTS ELEC. RM & NEW RING	L	0.00			201	11					R	-																																		
	WEST PRT - LIGHTING	L	0.30			201	13	14		1.20	1.20	R	-																																		
	WEST PRT - LIGHTING	L	0.30																																												

(E) PNL "NW" W(E) SCHED.

VOLTS: 208Y/120 PHASE: 3 WIRING: 4 BUSING: 250A POLES: 42		PANEL SCHEDULE EXIST NEW		MAIN BREAK: FED FROM: LOCATION: B116 SURFACE MOUNTED: 10AUG AIC RATING:								250A PNL EAST VA TOWNAUTPWR					
LOAD DESCRIPTION														TYPE		LOAD DESCRIPTION	
LIGHTING N.E. TOWER/MEZANINE														L		RECEPT - RING	
SPARE														-		RECEPT - RING	
SPARE														-		RECEPT - RING	
LIGHTING														L		RECEPT - 30A	
LIGHTING IC														L		RECEPT BY THE DOOR	
SPARE														-		SPACE	
RECEPT - IC ROOM 208														R		SPACE	
CONDENSER UNIT														L		SPACE	
CONDENSER UNIT														L		SPACE	
AL-2														H		SPACE	
RECEPT - IC ROOM														R		SPACE	
SPACE														-		SPACE	
SPACE														-		SPACE	
SPACE														-		SPACE	
SPACE														-		SPACE	
SPACE														-		SPACE	
TOTALS														7.25 7.49 6.32		7.18 6.54 6.54	
DRAWING # A08189														CONTR. FACTOR 1.00		LIVEST. KVA 15.76	
TYPE "N" NON-CONTINUOUS (MISC) LOADS														0.55		1.00 1.56	
TYPE "L" LIGHTING/CONTINUOUS LOADS														1.20		100% 1.56	
TYPE "N" RECEPTACLES (FIRST 100VA)														1.75		100% 16.00	
TYPE "N" RECEPTACLES (OVER 100VA)														0.00		100% 0.76	
TYPE "N" LARGEST MOTOR LOAD														14.87		120% 16.71	
TYPE "N" HVAC MECHANICAL LOADS														0.00		100% 0.00	
TOTALS														4.56		6.32	

AREA OF WORK

AREA OF WORK

117.48 AMPS

(E) PNL "NE" W(E) SCHED.

1

VOLTS:	208Y/120	EXIST	NEW	MAIN BREAK:	250A
PHASE:	3			FED FROM:	PNL WEST VA TOWNAUTPWR
WIRING:	4			LOCATION:	B112 SURFACE
BUSING:	250A			MOUNTED:	10AUG
POLES:	42			AIC RATING:	

PANEL SCHEDULE NW													
LOAD DESCRIPTION	TYPE	A	B	C	BRK	CKT	CKT	BRK	A	B	C	TYPE	LOAD DESCRIPTION
RING - LIGHTING	L	0.30			201	1	2	201	0.54			R	RING - RECEPT
RING - LIGHTING	L	0.30			201	3	4	201	0.54			R	RING - RECEPT
RING - LIGHTING	L	0.30			201	5	6	201	0.54			R	RING - RECEPT
RING - LIGHTING	L	0.30			201	7	8		1.20			R	
SPARE	-	0.00			201	9	10	300	1.20			R	RECEPT - 30A
EM. LIGHTS ELEC. RM & NEW RING	L	0.00			201	11	12		1.20			R	
WEST PRT - LIGHTING	L	0.30			201	13	14		1.20			R	
WEST PRT - LIGHTING	L	0.30			201	15	16	300	1.20			R	RECEPT - 30A
CONDENSER UNIT	L	0.00			201	17	18		1.20			R	
SPARE	-	0.00			202	19	20	201	0.00			-	SPARE
SPARE	-	0.00			201	21	22	201	0.72			R	VECTA SCANNER
RECEPT - TUNER ROOF TRAY	R	0.72			201	23	24	201	1.50			R	RACK POWER
FAN COIL "B116-FCL-1001"	H	0.00			201	25	26	201	1.00			R	RECEPT - 30A
RECEPT - 30A	R	0.00			201	27	28	202	1.20			R	
RECEPT - 30A	R	0.00			201	29	30		1.20			R	
RECEPT - 30A	R	0.00			201	31	32	201	0.00			-	SPARE
RECEPT - 30A	R	0.00			201	33	34	201	0.00			-	SPARE
RECEPT - 30A	R	0.00			201	35	36	201	0.00			-	SPARE
RECEPT - 30A	R	0.00			201	37	38	201	0.00			-	SPARE
RECEPT - 30A	R	0.00			201	39	40	201	0.00			-	SPARE
RECEPT - 30A	R	0.00			201	41	42	201	0.00			-	SPARE
TOTALS		4.42	0.00	0.00					4.42	0.00	0.00		

BOLD = NEW LOAD, NEW BREAKER

DRAWING #	ASB189	COUL	DEMAND FACTOR	DEMAND	DEMAND
EQUIP. TAG					
TYPE "N" NON-CONTINUOUS (MISC) LOADS	9.02	100%	6.00		
TYPE "L" LIGHTING/CONTINUOUS LOADS	0.00	100%	0.00		
TYPE "N" RECEPTACLES (FIRST 100VA)	0.00	100%	0.00		
TYPE "N" RECEPTACLES (OVER 100VA)	0.00	50%	0.00		
TYPE "N" LARGEST MOTOR LOAD	0.00	100%	0.00		
TYPE "N" HVAC MECHANICAL LOADS	27.00	100%	2.04		

PHASE A:	15.94	KVA
PHASE B:	16.00	KVA
PHASE C:	16.80	KVA

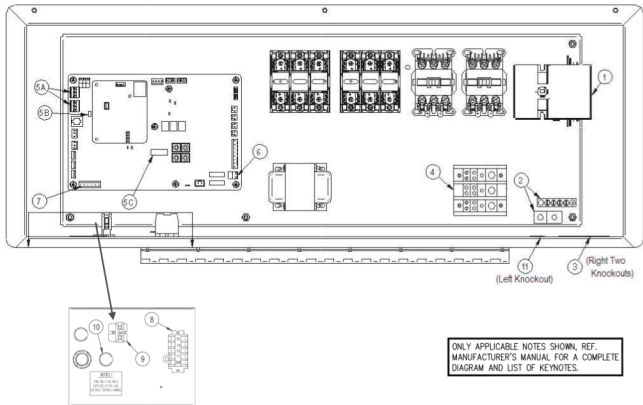
147.31 AMPS

(E) PNL "NW" W(N) SCHED.

VOLTS: 208Y/120		EXIST ☐ NEW ☐		MAIN BREAK: 250A		FED FROM: PNL EAST VA TOWNAUTPWR									
PHASE: 3						LOCATION: B116 SURFACE									
WIRING: 4						MOUNTED: 10AUG									
BUSING: 250A															
POLES: 42															
PANEL SCHEDULE NE															
NE															
LOAD DESCRIPTION	TYPE	A	B	C	BRK	EXT	CKT	BRKR	A B C TYPE LOAD DESCRIPTION						
LIGHTING N.E. EMER/EXITZANNE	L	0.30			201	1	2	201	0.54 R RECEPT - FENG						
SPACE					201	3	4	201	0.54 R RECEPT - FENG						
SPACE					0.00	1	6	201	0.54 R RECEPT - FENG						
LIGHTING	L	0.30			201	7	8	201	0.54 R RECEPT - FENG						
SPACE					0.00	9	16	300	1.20 R RECEPT - 30A						
LIGHTING IC	L	0.30			0.00	11	12		1.20 R RECEPT - 30A						
SPACE					0.00	13	14		1.20 R RECEPT - 30A						
RECEPT IC ROOM 208	R	0.54			302	15	16	303	1.50 R RECEPT - 30A						
SPACE					0.54	17			1.50 R RECEPT - 30A						
SPACE					0.50	20	22	300	1.20 R RECEPT - 30A						
SPACE					0.00	21	22		1.20 R RECEPT - 30A						
SPACE					0.00	23	24	300	1.20 R RECEPT - 30A						
FAN COIL "B116-FCU-1001"	M	8.00			1003	25	26	250	1.00 H BLU BACK B116						
RECEPT IC ROOM	R	0.54			0.54	29	30	201	1.50 R BLU BACK B116						
CONDENSER "B116-RCU-1001"	M	0.68			131	32	32	1.20	1.20 R RECEPT BY THE DOOR						
CONDENSER "B116-RCU-1001"	M	0.68			153	33	34		1.20 R RECEPT BY THE DOOR						
RECEP BY CONDENSER UNIT	R	0.18			0.68	35	36		0.36 R SPACE						
SPACE					0.18	37	38		0.36 R SPACE						
SPACE					0.18	39	40		0.36 R SPACE						
SPACE					0.18	41	42		0.36 R SPACE						
		10.40	10.52	11.00			7.18	6.94	6.94						
DRAWINGS #															
EQUIP TAGS															
ADDR300															
TYPE "N" NON-CONVENTIONAL MISC. LOADS		0.00	100%		0.00										
TYPE "L" LIGHTING CONTINUOUS LOADS		1.28	100%		1.50										
TYPE "M" MISC/SPAC (NOT PLUG) LOADS		0.76	100%		0.76										
TYPE "R" RECEPTACLES OVER 100VA		7.00	100%		5.00										
TYPE "H" LARGEST MOTOR LOAD		2.00	100%		2.00										
TYPE "F" HVAC MECHANICAL LOADS		2.00	100%		7.04										
TOTAL LOADS															
TOTAL LOADS															

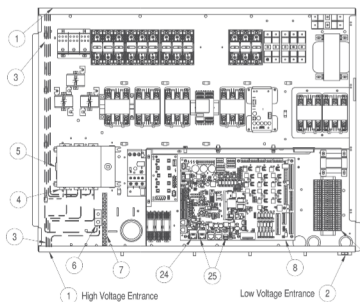
BOLD = NEW LOAD, NEW WIRING
PHASE A = 17.64 KVA
PHASE B = 17.36 KVA
PHASE C = 18.00 KVA
155.91 AMPS

REV	DESCRIPTION	DRN	CHK	APP	DATE
A	50% CD	HF	JO	SK	10-06-2025
B	100% CD	HF	JO	SK	11-06-2025
C	FAO REVIEW	HF	JO	SK	11-24-2025
D	BIO REVIEW	HF	JO	SK	12-18-2025



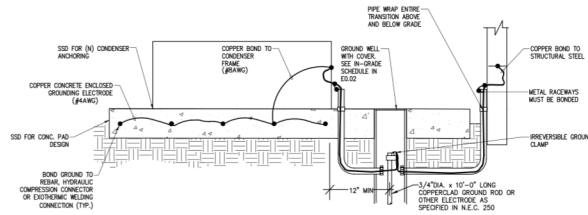
- Three phase electrical service – Terminals are on top of disconnect switch for one and two fan units. Terminals are on bottom of disconnect switch for three and four fan units. Three phase service not by Vertiv. See note 5.
- Primary high voltage entrance – Two 7/8" (22.2mm) diameter knockouts located at the bottom of the enclosure.
- Primary low voltage entrance – One 7/8" (22.2mm) diameter knockout that is free for customer low voltage wiring.

7 LIEBERT MC ELECTRICAL WIRING DIAGRAM
CONDENSER UNIT

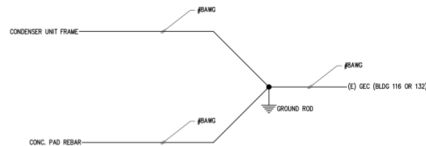


- High Voltage Entrance. Supplied on top and bottom of electric box. Knockout size Ø1.75in (44.5mm).
- Low Voltage Entrance. Ø1.375 in. (34.9mm) hole located on bottom of Electric Box.
- Three phase Electric Service and earth ground. Field supplied.
- Three phase connection. Electric service connection terminals on disconnect.
- Factory installed disconnect switch. Fused disconnect switch provided on units.
- Earth ground connection. Connection terminals for field supplied earth grounding wire.
- Earth ground bar. Connection terminals with factory ground from each high voltage component for field supplied earth grounding wire.

8 LIEBERT PDX ELECTRICAL WIRING DIAGRAM
FAN COIL UNIT

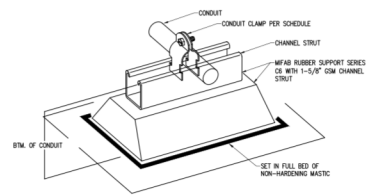


4 PAD MOUNTED
EQUIPMENT BONDING

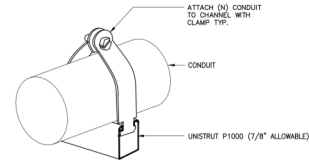


5 CONDENSER AND CONC. PAD REBAR GROUNDING

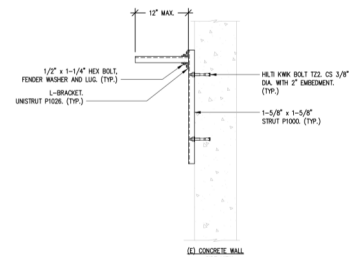
	SUPPORT OF CONDUIT		
	STD PIPE SIZE	CLAMP	CLAMP SPACING (CENTER TO CENTER)
HORIZONTAL	1/2 (0.840 O.D.)	P1111	5'-0"
	3/4 (1.050 O.D.)	P1112	6'-0"
	1 (1.315 O.D.)	P1113	6'-0"
	1 1/4 (1.660 O.D.)	P1114	6'-0"
	1 1/2 (1.900 O.D.)	P1115	6'-0"
	2 (2.375 O.D.)	P1117	6'-0"
VERTICAL	SEE ABOVE	SEE ABOVE	EVERY FLOOR LEVEL



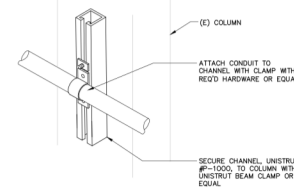
6 TYP. CONDUIT SLEEPER SUPPORT
ROOF SURFACE



1 TYP. CONDUIT SUPPORT DETAIL
CHANNEL STRUT



2 TYP. CHANNEL STRUT SUPPORT DETAIL
CONC. WALL



3 TYP. CONDUIT SUPPORTS
COLUMNS OR WALLS

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (FC)	
BY: ☐ order	
DATE: 10/2/2025	PRJ# 25-204
Revised Authorization	☒ No ☐ Yes
These plans have been reviewed and approved for construction. The contractor is responsible for obtaining all necessary permits and for compliance with all applicable codes and regulations. The authorization of these plans does not constitute a warranty or endorsement of any products, materials, or methods.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

All changes from the BIO approved job set of plans and specifications shall be submitted to, and approved by the BIO office prior to the changes being made in the field.

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2603 Camino Ramon, Suite 417
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ENGR: HENRI J. BERRY DES: PETER J. HARRIS CHK: JEFFREY L. HARRIS	DATE: 10/2/2025 APPROVALS: 10/2/2025 DATE: 10/2/2025	DRAWING NUMBER: B0116-E05-0015 REVISION: D	E

